



Smart Lending.
Stronger Decisions.

Final Live Presentation

Loan Default Prediction

From predictive model to business-ready credit-risk decision support

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Business challenge

Identify elevated-risk borrowers early enough to reduce credit losses - without unnecessarily restricting satisfactory borrowers.



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Resilience



Measurable
Business Impact



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Executive Summary

End-to-end solution, test evidence and business value

The project converts loan-default prediction into a practical, bank-ready credit-risk decision-support workflow.

Business problem

01

Manual underwriting can be slow, inconsistent and difficult to scale. The bank needs earlier risk triage without automatically rejecting good borrowers.

Selected solution

02

Tuned oversampled Gradient Boosting with a 0.34 enhanced-review threshold - not a rejection threshold.

Independent test evidence

03

PR-AUC 0.86 • Recall $\approx$ 82% • 194 of 238 historical defaults detected on unseen data.

Business impact

04

About 22% of applications are routed to enhanced review, allowing the credit team to focus attention where risk is concentrated.

Final recommendation: sell as a bank-specific pilot solution, validate on the bank's own portfolio, then scale with governance.



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Why this loan-default problem matters in the real market

A useful model should improve both credit-risk detection and the consistency of the loan-review process

Market signal: bad loans remain a banking concern

Non-performing loans (NPL) measure the share of bank loans classified as impaired or non-performing.

CBN prudential benchmark



Reported NPL ratio, Feb. 2026



Key drivers mentioned in Nigerian banking discussions



Source: CBN reports and Nigerian banking press, 2026

My business experience: why process design matters

As owner of Mifor Consult, I experienced two long loan-approval journeys where documentation was requested at different stages.

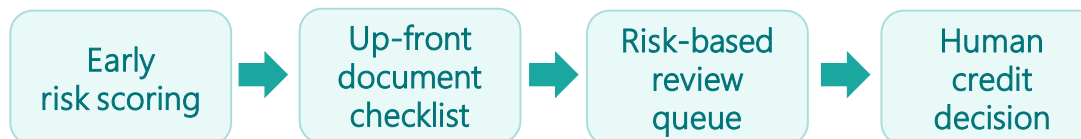
Fidelity Bank

Approval took more than one year

Zenith Bank

Approval took close to one year

Business lesson for this capstone



The model should reduce uncertainty early in the workflow - not bypass due diligence or replace credit officers.

What the Data Told Us Before Modelling

Early evidence translated into real-world credit-risk themes



Repayment capacity

Borrowers with weaker or missing affordability information showed higher historical default risk.



Adverse credit behaviour

Prior delinquencies and derogatory events were among the clearest warning signals.



Credit-history maturity

Shorter or thinner credit history was associated with greater uncertainty and risk.



Collateral and property information

Missing or weaker property/collateral evidence increased risk signals.



EDA hypothesis

Default risk is driven by the combined effect of repayment capacity, prior credit behaviour, information quality and credit-history depth.



From Hypothesis to Predictive Model

A leakage-safe modelling process designed for credible business decisions

Raw applications

Historical approved-loan portfolio

Train / validation / test

Independent final holdout

Feature engineering

Business-driven risk variables

Preprocessing

Leakage-safe transformations

Candidate models

5 algorithms compared

Tuning + threshold

Validation-only decisions

Independent test

Final generalisation evidence

A Models evaluated

Logistic Regression • KNN
Decision Tree • Random Forest
Gradient Boosting

B Imbalance strategies

Original distribution •
Oversampling
Undersampling • Class weighting

C Selection logic

Choose the model that balances default detection, false alarms, ranking quality, probability reliability and business usability.

Implementation mindset

Build only what management can verify, explain, pilot and monitor.



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Why Gradient Boosting Won

Final choice based on business trade-off, not accuracy alone



SELECTED

Tuned Oversampled
Gradient Boosting

0.90 **0.83** **0.81**
PR-AUC Recall Precision
Brier 0.06

Balanced alternative

Tuned Original
Random Forest

0.87 **0.76** **0.84**
PR-AUC Recall Precision
Brier 0.06

Maximum recall

Undersampled
Random Forest

0.79 **0.89** **0.65**
PR-AUC Recall Precision
Brier 0.10

Selection principle

Best overall trade-off between detecting defaulters, controlling false alarms, ranking borrowers reliably and supporting practical credit review.



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Why the Threshold Is 0.34, Not 0.50

The selected threshold routes applicants to enhanced review - not automatic rejection

Selected threshold: 0.34

206 defaults detected
62 satisfactory applicants reviewed
F2-score: 0.84

Standard threshold: 0.50

197 defaults detected
45 satisfactory applicants reviewed
F2-score: 0.82



+9 additional defaults detected for +17 additional satisfactory reviews

0.34 = intervention threshold, not a rejection rule

Did It Generalise?

Final evidence from unseen test applications

82%

of historical test defaults
identified

Locked model • threshold 0.34

0.86

PR-AUC

0.95

ROC-AUC

Confusion-matrix meaning

194

Defaults
detected

44

Defaults missed

72

Good borrowers flagged

882

Good retained

Risk bands: realised default rates rise sharply

Default risk is strongly concentrated in the very-high-risk segment.

Lower risk



3.41%

Moderate risk



30.43%

High risk



44.07%

Very high risk



81.16%



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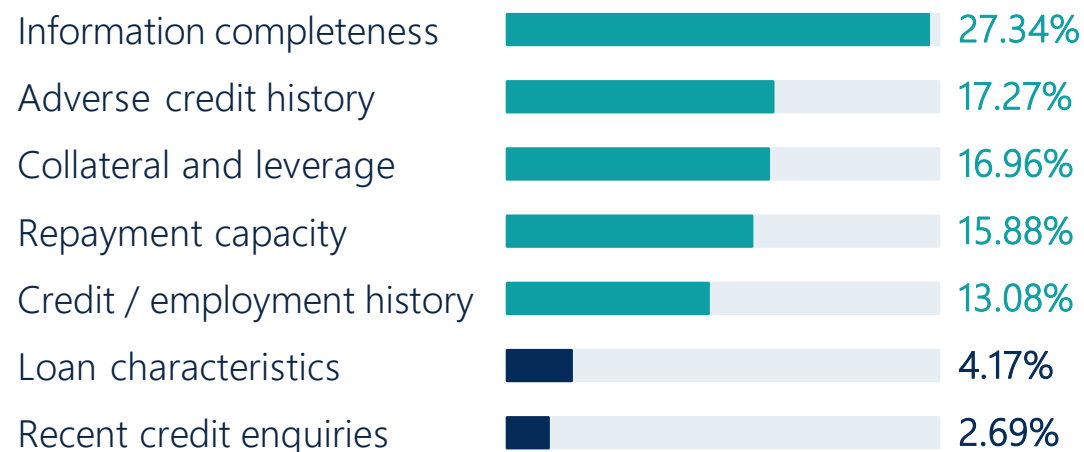


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What Actually Drives Default Risk?

EDA findings and the final model tell a consistent business story

Business-level risk drivers



EDA suggested

Incomplete affordability data + adverse credit history + thin credit history = elevated risk.

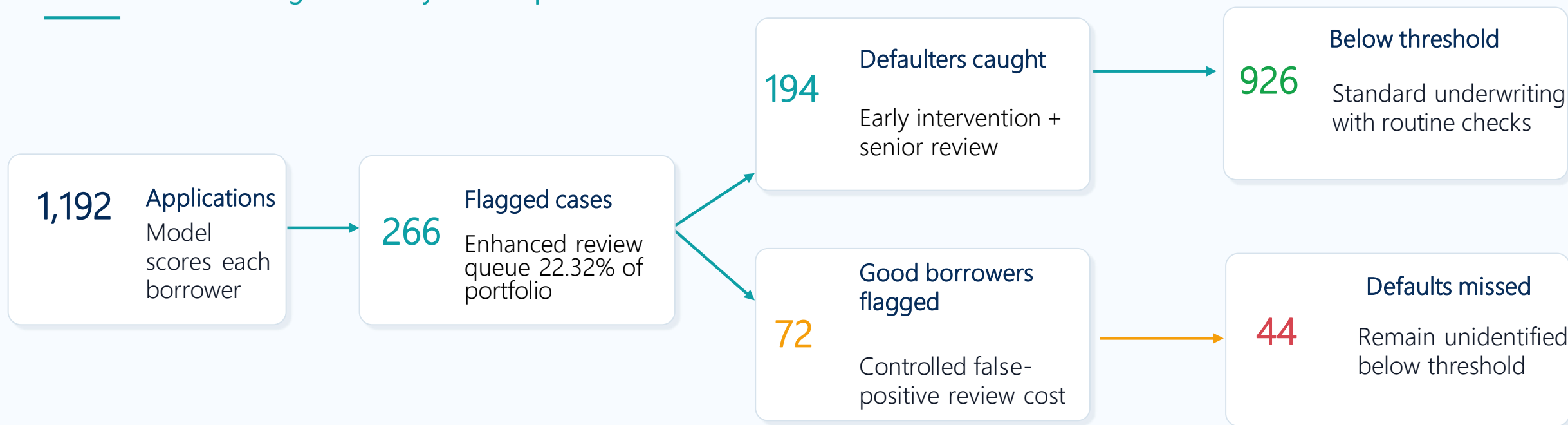
Final model confirmed

The strongest predictive themes are information completeness, credit behaviour, collateral/leverage and repayment capacity.

Management message: verify repayment capacity and missing financial information before making credit decisions.

A Lending Manager Receives 1,192 Applications

How the model changes the daily review queue



Management implication

Focus additional credit resources on about one-fifth of applications while capturing about four-fifths of historical defaults.



Business Impact

How model performance translates into portfolio, operational and financial value



82%

Risk detection

Historical test defaults identified: 194 of 238.



22%

Focused review

Only the highest-risk 266 of 1,192 applications are routed to enhanced review.



≈81%

Exposure proxy captured

3.51m of 4.35m default-related approved-loan exposure proxy identified.



Faster

Operational efficiency

Credit officers focus intensive effort on complex/high-risk files.

Illustrative financial impact

878,600

potential avoided loss under central scenario

Assumption: 50% LGD × 50% intervention effectiveness.

Main business value

Protect portfolio value • prioritise scarce review resources • improve underwriting consistency • identify elevated risk earlier.

Not a replacement for KYC, collateral checks, compliance or final human credit decisions.

Business impact: better prediction becomes valuable only when it changes review effort, risk control and management decisions.



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Commercial Value, Efficiency & Bank Adoption

Can this idea be sold to a bank - and what value would it create?

Financial & risk value



- Earlier identification of likely defaulters
- Reduced avoidable credit-loss exposure
- Better protection of interest income and portfolio quality
- Evidence-based prioritisation of intervention resources

Operational efficiency



- First-pass risk assessment once key data are captured
- About 22% routed to enhanced review in the test scenario
- Senior credit officers focus on complex and risky cases
- Improves consistency and application-processing capacity

Commercial potential



- Sellable as a bank-specific decision-support system
- Can be delivered as dashboard, API or workflow tool
- Requires bank-specific recalibration and validation
- Value proven through pilot outcomes, not claims alone

Commercial proposition: Faster risk triage + consistent underwriting + focused human review + earlier default-risk detection

Important: the model supports credit officers; it does not replace KYC, collateral verification, compliance checks or final human lending decisions.



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Business Recommendations

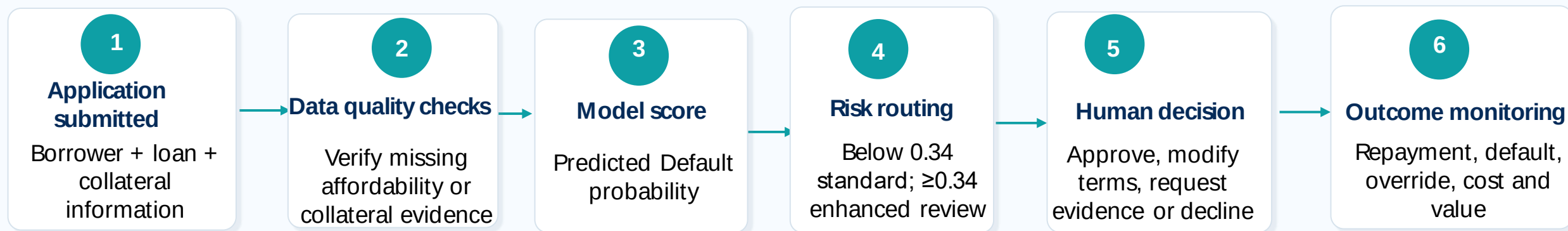
Practical actions management should take from the model

- 1** Pilot before scale
Run a controlled pilot and measure live outcomes before full production deployment.
- 2** Use 0.34 for review
Route cases above the threshold to enhanced review, not automatic rejection.
- 3** Improve docs upfront
Use a clear checklist for affordability, collateral and missing financial information.
- 4** Create risk-based queues
Prioritise high-risk cases for senior review; route lower-risk cases normally.
- 5** Monitor value and fairness
Track defaults caught, false positives, overrides, review cost, borrower groups and drift.

Pilot → measure → govern → scale responsibly

How the Lender Would Actually Use It

A practical, human-supervised workflow informed by business delivery experience



Operational rule: The model prioritises human attention. It does not replace underwriting judgement.
Mifor Consult business lens: technical insight → verification → management decision → documented action → outcome monitoring.

Controls needed before full deployment

A real-world deployment plan must combine model evidence, business controls, fairness and monitoring.



Governance

Monitor calibration, drift, recall, precision, risk-band stability and model overrides.



Fairness & bias control

Do not use employment or missing information as automatic exclusion rules; monitor borrower-group outcomes.



Management decision

Proceed to a controlled pilot before full production deployment. Pilot, measure, govern, then scale.

Final conclusion Mifor Consult perspective: pilot first, measure real outcomes, document decisions and scale responsibly.



Conclusion

Final business message from the end-to-end solution

The model helps lenders detect risk earlier, focus review, and govern decisions better.

82% Defaults identified

Strong test-set default capture.

22% Review queue

Focus effort on the riskiest segment.

Pilot Next step

Measure value and fairness before scale.

Final recommendation: controlled operational pilot before full production deployment.



Thank You

Questions and discussion

Predict risk earlier.
Focus review where it matters.
Protect portfolio value.

Recommended next step: controlled pilot with monitored outcomes, human oversight and business-governance discipline.

